ENEE12015 - Electrical Power Engineering

Lecture Notes

Authored by
Lachlan J. W. Wyer

Document Start Date: 5th of November 2026
Latest Update: November 12, 2026

Contents

1 Introduction to Electrical Power Systems **4**
- 1.1 Early Development: . . . 4
- 1.2 Power System Compenets: . . . 4
- 1.3 Power System Models: . . . 4
  - 1.3.1 Generators . . . 4
  - 1.3.2 Transmission/Distribution Lines . . . 5
  - 1.3.3 Transformers . . . 5
  - 1.3.4 Loads . . . 5
- 1.4 Basic Theory: . . . 5
- 1.5 Current and Power Flow: . . . 6
- 1.6 Power in AC Circuits: . . . 6
- 1.7 Reactive Power: . . . 6
- 1.8 Power Triangles: . . . 7
- 1.9 Power Flow Direction: . . . 7

2 Three Phase Power **7**
- 2.1 What is it?: . . . 7
- 2.2 Balanced Operating Conditions: . . . 7
- 2.3 Analysis of 3-Phase: . . . 7
- 2.4 Popularity of Polyphase Systems: . . . 8
- 2.5 Power in Polyphase Systems: . . . 8
- 2.6 3-Phase Connections: . . . 10
- 2.7 Balanced Systems: . . . 11
- 2.8 3-Phase Device Connection: . . . 11
- 2.9 Phase Sequence: . . . 12
- 2.10 3-Phase Solutions: . . . 12
- 2.11 Symmetrical Component Analysis: . . . 12
- 2.12 Unbalanced Decomposition: . . . 12
- 2.13 Balanced Analysis: . . . 13
  - 2.13.1 Wye Connection . . . 13
  - 2.13.2 Delta Connection . . . 13
  - 2.13.3 All . . . 13
- 2.14 Selecting Number of Phases: . . . 14

3 Transformers **14**
- 3.1 Voltage Induced in a Coil: . . . 14
- 3.2 Using Voltage: . . . 15
- 3.3 Core Types: . . . 15
- 3.4 Elementray Transformers: . . . 15

3.5 Ideal Transformer: . . . 16
3.6 Ideal Under Load: . . . 16
3.7 Impedance Transformation: . . . 17
3.7.1 Secondary to Primary . . . 17
3.8 Actual Transformer Core: . . . 18
3.9 Leakage Flux: . . . 19
3.10 Complete Equivalent Circuit: . . . 19
3.11 Simplifing Circuit: . . . 19
3.11.1 No Load . . . 19
3.11.2 Full Load . . . 20
3.12 Measuring Transformer Impedances: . . . 21
3.13 Terminal Windings: . . . 22

4 Per Unit Systems **23**
4.1 Introduction: . . . 23
4.2 Single Phase Systems: . . . 23
4.3 Converting to PU Values: . . . 24
4.4 Further Simplification: . . . 25
4.5 3-Phase Systems: . . . 25
4.6 Advanced Concepts: . . . 26

5 3-Phase Transformer **26**
5.1 Dual Voltage Transformer: . . . 26
5.2 Auto Transformer: . . . 26
5.3 Conventional Transformer as an Auto Transformer: . . . 27
5.4 Voltage Transformer: . . . 28
5.5 Current Transformers: . . . 28
5.6 Toroidal Current Transformer: . . . 29
5.7 Variable Auto-Transformer: . . . 29
5.8 Induction Heating Transformer: . . . 29
5.9 High Impedance Transformer: . . . 29
5.10 High Frequency Transformer: . . . 30
5.11 3-Phase Transformers Core Config: . . . 30
5.12 3-Phase Connections: . . . 30
5.12.1 Delta-Delta . . . 31
5.12.2 Delta-Wye . . . 31
5.12.3 Wye-Wye . . . 32
5.12.4 Open-Delta Connection . . . 32
5.13 Vector Group for Transformer Connections: . . . 32
5.14 ANSI Standard Connection: . . . 33
5.15 Special 3-Phase Auto Transformers: . . . 33

5.16 Phase Shift Principle: 34
5.17 Analyzing 3-Phase Transformers within Circuits: 34

1 Introduction to Electrical Power Systems

1.1 Early Development:

1800's - Electricity and Magnetism well understood intellectually
1831 - Magnetica field was discovered to generate electricity
1845 - First commerical operations of telegraph
1860 - First electrical generator
1870 - First incandesent light bulb
1882 - First commericial power system in NY City
1883 - Polyphase induction motor
1885 - Commericial availability of AC power

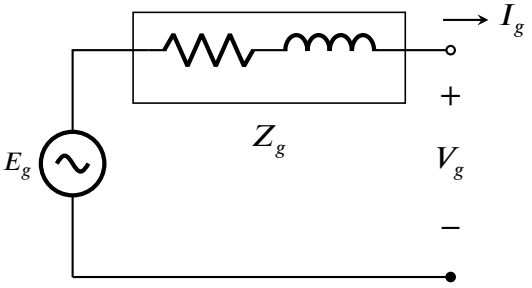
1.2 Power System Compenets:

- Generators
- Transformers
- Transmission lines
- Substations/Terminal Stations
- Distribution lines
- Electrical loads

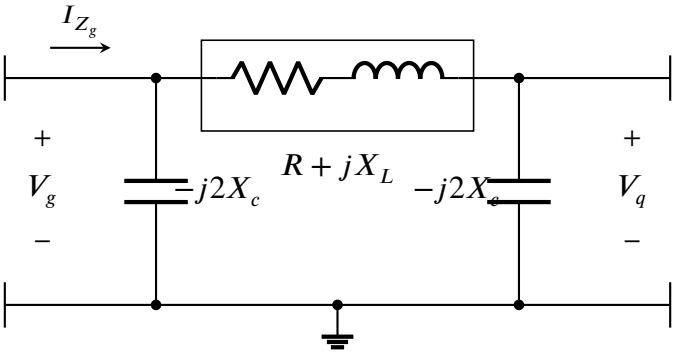
When transmitting power, the best senario is high voltage, low current transmission. Ideally below the resistivity of the power lines.

1.3 Power System Models:

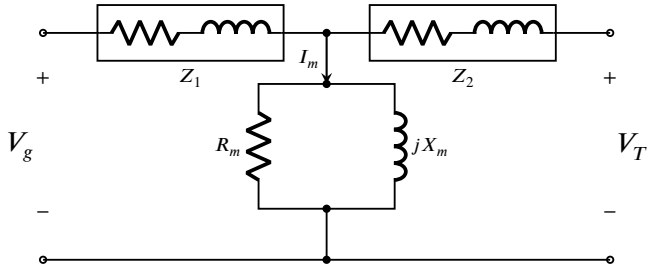
1.3.1 Generators



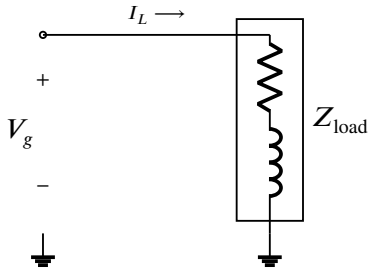
1.3.2 Transmission/Distribution Lines



1.3.3 Transformers



1.3.4 Loads



1.4 Basic Theory:

Ohm's Law - $V = IZ$
KVL - $\sum V = 0$

KCL - $\sum I_{in} = \sum I_{out}$
Power Flow - $S = P + jQ = VI^*$

1.5 Current and Power Flow:

- Technically current flow within an AC system is arbitrary
- Power is clear from source to load
- Steady state is only assumed

1.6 Power in AC Circuits:

Average Power over time period T

$$P = \frac{1}{T}\int_{t_0}^{t_0+T} v(t)i(t)dt$$

Instantaneous Power

$$p(t) = v(t)i(t)$$

When a circuit is in steady state, all voltages and currents become sinusodial. The periodic nature of the waveforms enables more specific derivation of power relationships.

$$P_{av} = \frac{E_m I_m}{2} = \frac{E_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} = E_{RMS} I_{RMS}$$

This only works when $v(t)$ and $i(t)$ are in phase. When a small phase shift occurs:

$$P_{av} = \frac{E_m I_m}{2}\cos(\theta) = E_{RMS} I_{RMS}\cos(\theta)$$

When $i(t)$ lags $v(t)$ by 90°

$$P_{av} = E_{RMS} I_{RMS}\cos(90°) = 0$$

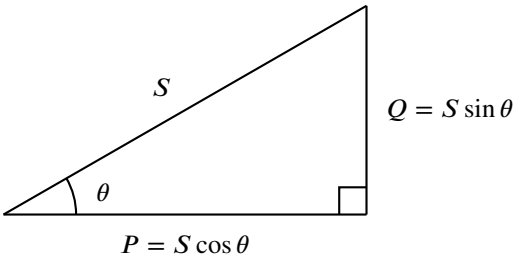
1.7 Reactive Power:

Power that dose no useful work within a system.

$$Q = E_{RMS} I_{RMS}\sin(\phi))$$

VAR is a defined value and does not relate to the area under P. When $v(t0$ and $i(t)$ are in phase ($\phi = 0$), $Q = 0$. When reactive power is maximum ($\phi = 90°$), $Q = E_{RMS} I_{RMS}$ and $P_{av} = 0$.

1.8 Power Triangles:



Q is wished to be minised, allowing the maximum S to used within the load (active power) usefully.

1.9 Power Flow Direction:

A positive P present in a power triangle indicates that ϕ is between $-90°$ and $90°$. With a negative P show ϕ to be $> 90°$. It is necessary to assign a current direction in AC circuits, for the purpose of writing equations in a consistent manner. $Q > 0$, flowing in the same direction as current is assumed to go. $Q < 0$, flowing apposed to the assumed current direction.
Power Factor:

$$pf = \frac{P}{S} = \cos(\phi)$$

2 Three Phase Power

2.1 What is it?:

A 3-phase power system is just a connecting of 3 singlephase systems together for set operation. Each phase must operate at the same voltage and current magnitudes seperated by $120°$ phase angle.

2.2 Balanced Operating Conditions:

Under normal operating conditions all voltages, current, $120°$ angle shift and loads contain same characteristics.

2.3 Analysis of 3-Phase:

Assuming a system is under balanced conditions, means that only a single phase system needs to be analysed. Faults, abnormal load conditions, cause a system

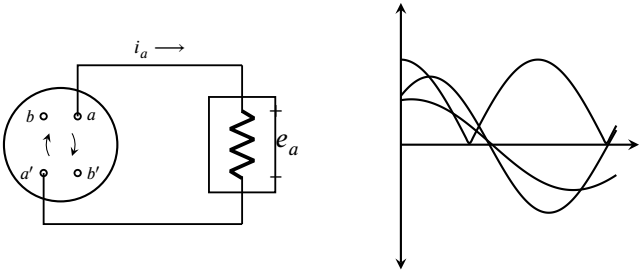
to become unbalanced.

2.4 Popularity of Polyphase Systems:

Polyphase machines can be designed to operate under constant total power conditions. Hence, these machines have smoother operations. These machines are then simpler to design, how all as tansmission lines can delivers greater power for less cost.

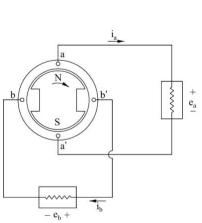
2.5 Power in Polyphase Systems:

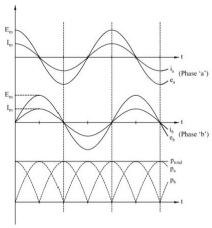
Single Phase: Resistive



$$
\begin{aligned}
p_a &= e_a i_a \\
&= E_m I_m \cos^2(\omega t) \\
P_{av} &= E_{RMS} I_{RMS}
\end{aligned}
$$

2-Phase: Resistive





Voltage induced in b-b' lags voltage in a-a' by 90 electrical degrees. In 2-pole machines, 1 electrical degree is equivalent to 1 mechanical degree.

$$\begin{aligned} e_a &= E_m \cos(\omega t) \\ i_a &= I_m \cos(\omega t) \end{aligned}$$

$$\begin{aligned} e_b &= E_m \cos(\omega t - 90°) \\ &= E_m \sin(\omega t) \\ i_b &= I_m \cos(\omega t - 90°) \\ &= I_m \sin(\omega t) \end{aligned}$$

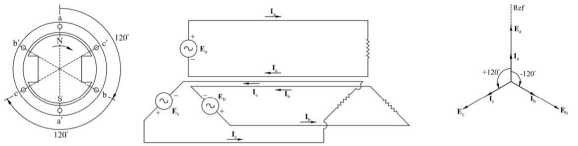
$$\begin{aligned} P_{Tot} &= P_a + P_b \\ &= E_m I_m \cos^2(\omega t) + E_m I_m \sin^2(\omega t) \\ &= E_m I_m \end{aligned}$$

There are two seperate loads. This P_{Tot} is true as long as the load is the same.

$$\begin{aligned} P_{av} &= P_{Tot} \\ &= 2E_{RMS} I_{RMS} \end{aligned}$$

$$\begin{aligned} S_{Tot} &= S_a + S_b \\ &= E_a I_a^* + E_b I_b^* \\ &= 2EI\angle 0° \end{aligned}$$

3-Phase: Resistive



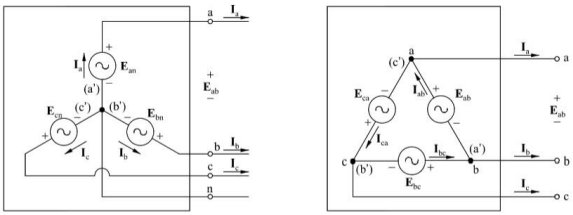
$$
\begin{aligned}
e_a &= E_m \cos(\omega t) \\
i_a &= I_m \cos(\omega t)
\end{aligned}
$$

$$
\begin{aligned}
e_b &= E_m \cos(\omega t - 120°) \\
i_b &= I_m \cos(\omega t - 120°)
\end{aligned}
$$

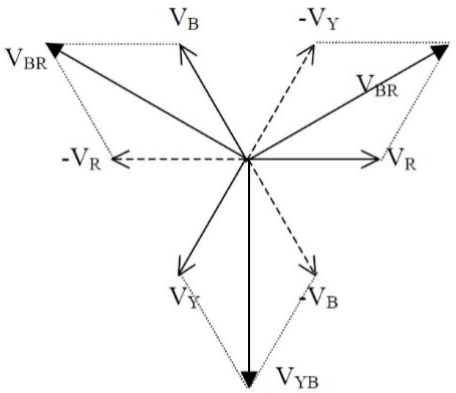
$$
\begin{aligned}
e_c &= E_m \cos(\omega t - 240°) \\
&= E_m \cos(\omega t + 120°) \\
i_c &= I_m \cos(\omega t - 240°) \\
&= I_m \cos(\omega t + 120°)
\end{aligned}
$$

$$
\begin{aligned}
P_{Tot} &= P_a + P_b + P_c \\
&= 1.5 E_m I_m \\
&= 3 E_{RMS} I_{RMS}
\end{aligned}
$$

2.6 3-Phase Connections:



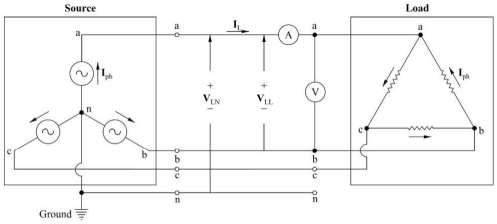
2.7 Balanced Systems:



- Angles between phases is 120°
- Magnitudes are all equal
- V_L would be 30° leading nearest phase voltage
- $|V_L|$ is $\sqrt{3}|V_P|$

$$I_L = I_P$$
$$|V_L| = \sqrt{3}|V_P|$$
$$|I_L| = \sqrt{3}|I_P|$$

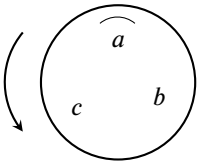
2.8 3-Phase Device Connection:



The delta-star and star-delta conversions can be done. The only requirement is that the device voltage ratings must be compatitable.

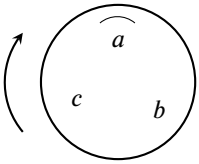
2.9 Phase Sequence:

The phase sequence denotes which phase becomes succesively positive.



+ve seq

$-a-b-c-a-b-c-$



–ve seq

$-a-c-b-a-c-b-$

2.10 3-Phase Solutions:

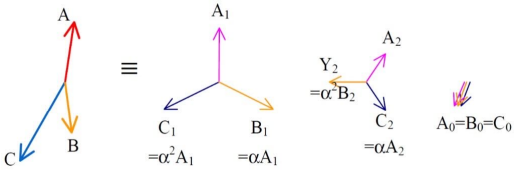
- Most 3-Phase system can be though of as balanced
- To solve an unbalanced 3-phase system is quite complex

2.11 Symmetrical Component Analysis:

Any unbalanced system can be split up into a series of balanced systems.

- Same phase sequence (Positive)
- Opposite phase sequence (Negative)
- In phase quantities (Zero)

2.12 Unbalanced Decomposition:

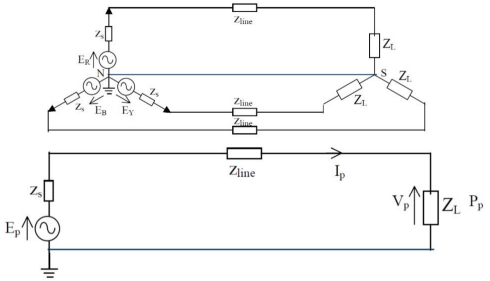


$$\alpha = 1\angle 120^\circ$$

$$A = A_0 + A_1 + A_2 = A_0 + A_1 + A_2$$
$$B = B_0 + B_1 + B_2 = A_0 + \alpha^2 A_1 + \alpha A_2$$
$$C = C_0 + C_1 + C_2 = A_0 + \alpha A_1 + \alpha^2 A_2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \end{bmatrix}$$

2.13 Balanced Analysis:



2.13.1 Wye Connection

$$I_P = I_L$$
$$V_P = \frac{V_L}{\sqrt{3}}$$

2.13.2 Delta Connection

$$V_P = V_L$$
$$I_P = \frac{I_L}{\sqrt{3}}$$

2.13.3 All

$$S_{3P} = 3V_P I_P = \sqrt{3} V_L I_L$$
$$P_{1P} = \frac{P_{3P}}{3}$$

2.14 Selecting Number of Phases:

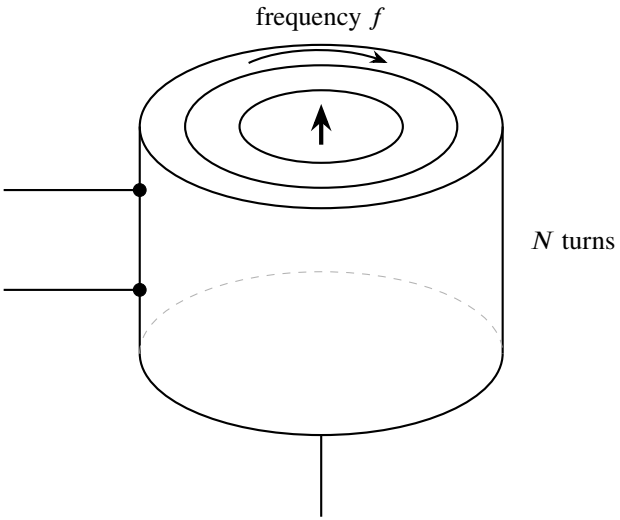
The number of phases for a power system is determined by transmission system economies. That is done by analyzing the amount of power transferred per conductor. With the assumption that the size of the return conductor is equal to that phase conductor.

System	Power Rating	No. Conductors	Power per Conductor
1-Phase	P	2	0.5P
2-Phase	2P	3	0.667P
3-Phase	3P	4	0.75P
4-Phase	4P	5	0.8P

3 Transformers

3.1 Voltage Induced in a Coil:

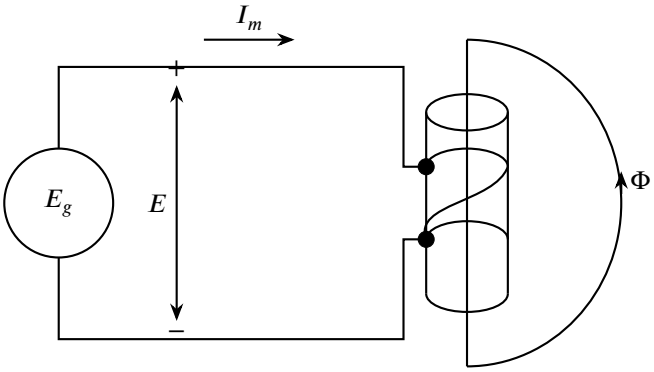
A voltage is induced in a coil when it links to a variable flux. The variable flux then induces a variable voltage.



$$E = 4.44 f N \phi_{max}, \text{ Where:}$$

- E - Effective induced voltage (V)
- f - Frequency (Hz)
- N - Number of turns within the coil
- ϕ_{max} - Peak flux value (Wb)

3.2 Using Voltage:



If the coil resistence is negligible, reactance is:

$$X_m = \frac{E_g}{I_m}$$

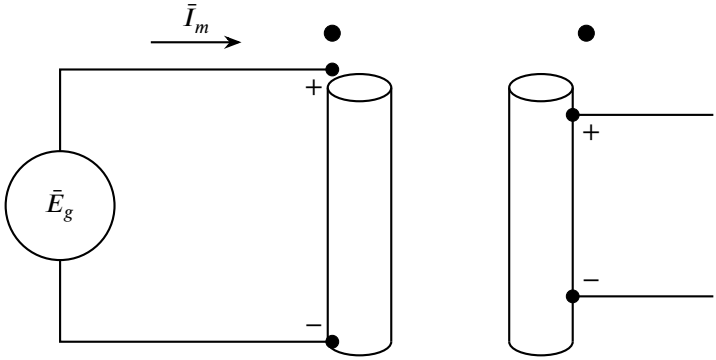
$$\phi_{max} = \frac{E_g}{4.44 f N}$$

3.3 Core Types:

The most common are air and iron. For the same E_g, the ϕ_{max} will be the same. The only difference is that the iron core requires less current.

3.4 Elementray Transformers:

The ϕ_{max1} can induce voltage E_{g2}. This is done throuhgh multual (ϕ_{m1} and leakage (ϕ_{f1}) flux. Dot notation is used to know the polarity of the induced voltage.



A current entering a polarity marked terminal produces a flux in a positive direction.

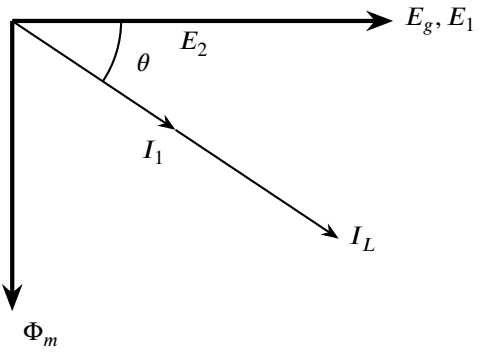
3.5 Ideal Transformer:

- An ideal transformer has no losses in its core, as it is infinitely permeable
- Fluxes are completely mutual with no leakage
- Polarity induced depends on the winding direction relative to the primary

$$E_1 = 4.44 f N_1 \phi_{max}$$
$$E_2 = 4.44 f N_2 \phi_{max}$$
$$a = \frac{E_1}{E_2} = \frac{N_1}{N_2}$$

3.6 Ideal Under Load:

- The mutual flux remains unchanged

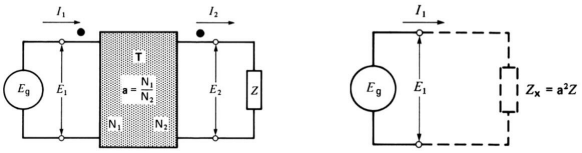


- ϕ_m can remain the same only if the primary side develops an mmf, which is counter-balanced by the secondary mmf, at all times

$$N_1 I_1 = N_2 I_2$$
$$a = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

3.7 Impedance Transformation:

3.7.1 Secondary to Primary

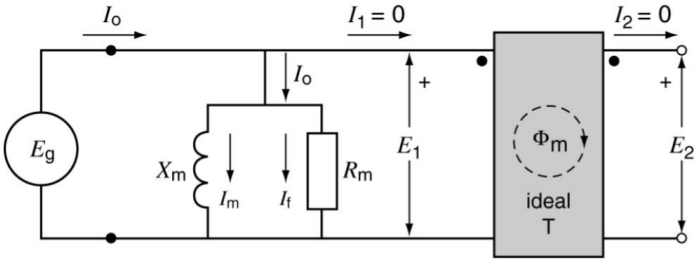


The effective impedance (Z_x) seen by E_g is:

$$
\begin{aligned}
Z_x &= \frac{E_1}{I_1} \\
Z &= \frac{E_2}{I_2} \\
\therefore Z_x &= \frac{E_1}{I_1} = \frac{aE_2}{\frac{I_2}{a}} \\
&= \frac{a^2 E_2}{I_2} \\
&= a^2 Z
\end{aligned}
$$

3.8 Actual Transformer Core:

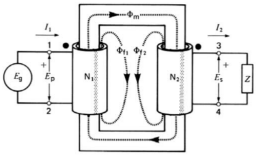
- Not infinitely permeable
- Has hysteresis loss
- Has eddy current loss
  - R_m - Iron core loss
  - X_m - Magnetizing reactance
  - Measure of permeability of transformer core
- R_m draws a small current I_f
- X_m current is I_m
- $E_g = E_1$, I_m lags E_1 by 90°



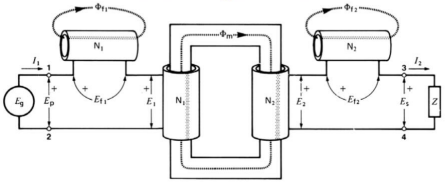
$$R_m = \frac{E_1^2}{P_m}$$

$$X_m = \frac{E_1^2}{Q_m}$$

3.9 Leakage Flux:

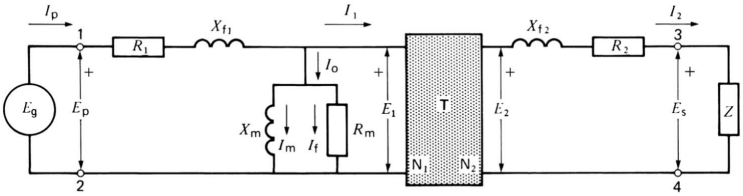


- Separating the various induced voltages due to the mutual flux and the leakage fluxes



Leakage flux occurs due to an imperfect coupling between coils.

3.10 Complete Equivalent Circuit:

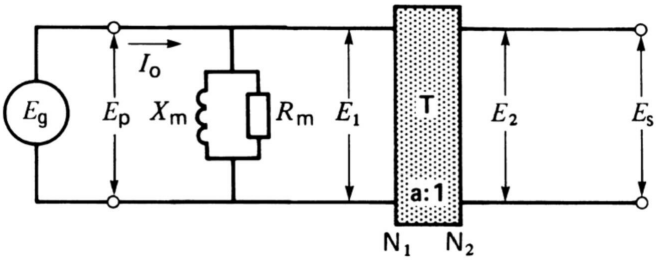


3.11 Simplifing Circuit:

3.11.1 No Load

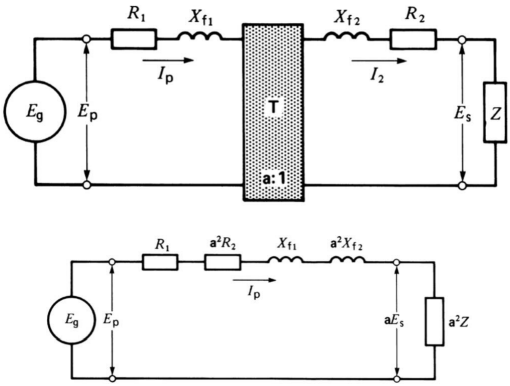
- $I_2 = 0$, as such current $I_1 = 0$

- Since, only I_o flows through R and X_{f1}, voltage drops across them can be neglected
- Turns raio can be found from the ratio of primary voltage and O/C secondary voltage



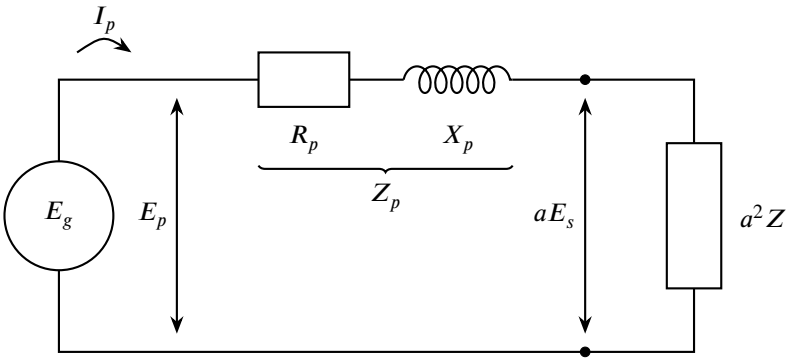
3.11.2 Full Load

- I_p will be approximately 20 time I_o
- Hence, I_o can be ignored



$$R_p = R_1 + a^2 R_2$$

$$X_p = X_{f1} + a^2 X_{f2}$$



3.12 Measuring Transformer Impedances:

O/C tests and determination of R_m, X_m and a within the core.

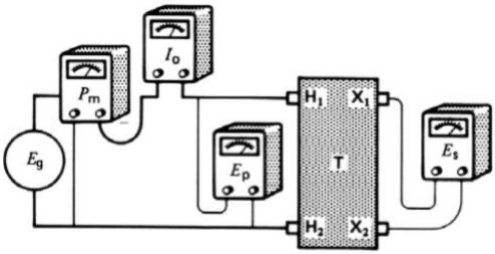
$$S_m = E_p I_o$$

$$Q_m = \sqrt{S_m{}^2 - P_m{}^2}$$

$$R_m = \frac{E_p{}^2}{P_m}$$

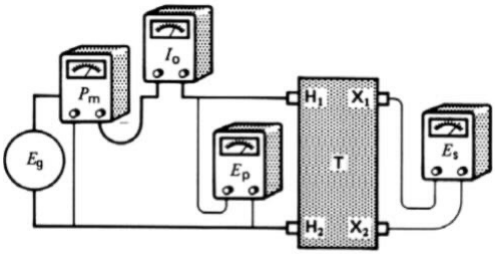
$$X_m = \frac{E_p{}^2}{Q_m}$$

$$a = \frac{E_p}{E_s}$$



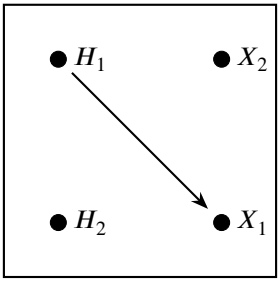
S/C test determines leakage reactance and winding resistance. The applied voltage for S/C testing must be <5% of rated.

$$Z_p = \frac{E_{SC}}{I_{SC}}$$
$$R_p = \frac{P_{SC}}{{I_{SC}}^2}$$
$$X_p = \sqrt{{X_p}^2 - {R_p}^2}$$

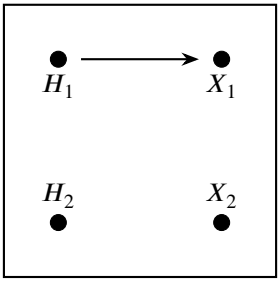


3.13 Terminal Windings:

- H_1, H_2 for the high voltage winding
- X_1, X_2 is the low voltage winding



Additive Polarity



Subtractive Polarity

Subtractive polarity is standard for all single-phase transformers above 200kVA provided the HV winding is 8660V. Most others are additive polarity.

4 Per Unit Systems

4.1 Introduction:

The PU values give a vibe about all quantities within a transformer. They are a unitless measurement. PU impedance values fall within a narrow range, and is somewhat independent of equipment ratings. This make erroneous data conspicuous. PU voltages are often better indicators of voltage levels. The PU system eliminates ideal transformers as circuit elements. This simplifies the transformers model and system solution.

4.2 Single Phase Systems:

- Select an arbitrary apparent power base for the whole system (S_{base}). A better choice would be to pick the power rating of the largest generator or source.
- Select a voltage base (V_{base}) for one of the circuits. Typically the rated voltage in the source circuit is used as the starting point.
- Establish V_{base} in other circuits based on transformer turn ratios.
- Find base current:

$$I_{base} = \frac{S_{base}}{V_{base}}$$

- Find base impedance:

$$Z_{base} = \frac{V_{base}}{I_{base}} = \frac{{V_{base}}^2}{S_{base}}$$

- Establish PU values for all quantities:

$$V_{pu} = \frac{V_{act}}{V_{base}}$$
$$I_{pu} = \frac{I_{act}}{I_{base}}$$
$$Z_{pu} = \frac{Z_{act}}{Z_{base}}$$
$$S_{pu} = \frac{S_{act}}{S_{base}}$$

- Consider act Z:

$$Z_{pu} = \frac{Z}{Z_{base}} = \frac{R + jX}{Z_{base}} = \frac{R}{Z_{base}} + j\frac{X}{Z_{base}} = R_{pu} + jX_{pu}$$

$$\therefore R_{base} = X_{base} = Z_{base}$$

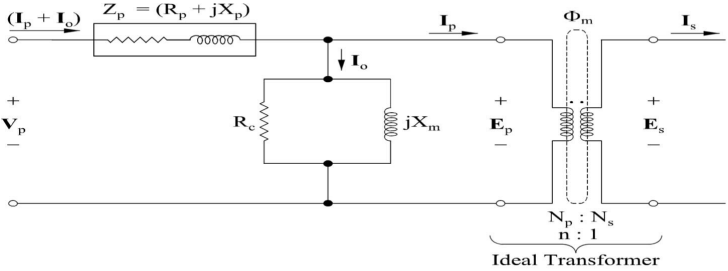
- Consider act S:

$$S_{pu} = \frac{S}{S_{base}} = \frac{P + jQ}{S_{base}} = \frac{P}{S_{base}} + j\frac{Q}{S_{base}} = P_{pu} + jQ_{pu}$$

$$\therefore P_{base} = Q_{base} = S_{base}$$

- Using all the values, solve the circuit

4.3 Converting to PU Values:



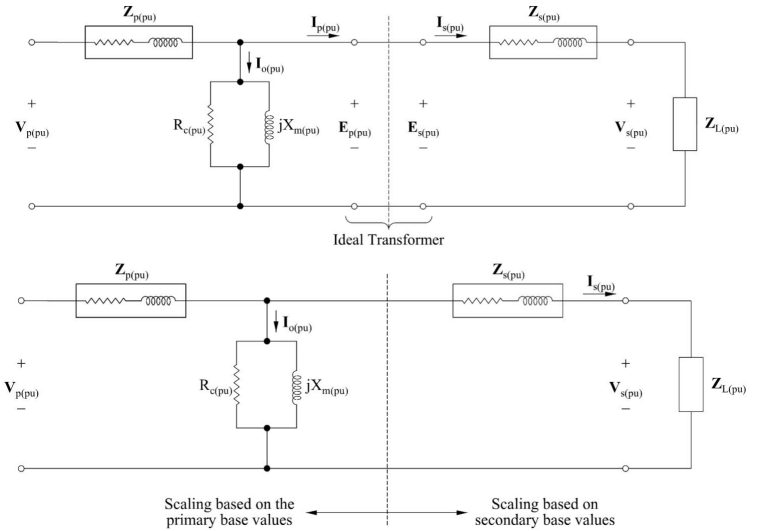
$$V_{s(base)} = \frac{V_{p(base)}}{n}$$

$$E_{p(pu)} = \frac{E_p}{V_{p(base)}}$$

$$E_{s(pu)} = \frac{E_s}{V_{s(base)}}$$

$$= E_{p(pu)}$$

$$I_{p(pu)} = I_{s(pu)}$$



4.4 Further Simplification:

Value of R_c and X_m are much large (by magnitude of 1000) compared to Z_p and Z_s. As such it is common to ignore the shunt branch.

4.5 3-Phase Systems:

- The procedure and conceps used in single phase are still valid. The only difference is in V_{LL}.
- All connected devices are assumed to be star connected, with all delta to be converted to the star equivalents.
- Select the apparent power base as the 3-phase rating (S_{3p}).
- Voltage bases must be defined:

$$V_{l(base)}$$

$$V_{p(base)} = \frac{V_{l(base)}}{\sqrt{3}}$$

- Line current:

$$I_{l(base)} = \frac{S_{3p}}{\sqrt{3}V_{l(base)}}$$

$$I_{p(base)} = I_{l(base)}$$

- Assuming star connected impedances are defined as:

$$Z_{base} = \frac{V_{p(base)}}{I_{p(base)}} = \frac{V_{l(base)}}{\sqrt{3}I_{l(base)}} = \frac{{V_{l(base)}}^2}{S_{3p}}$$

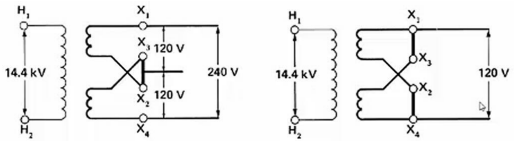
4.6 Advanced Concepts:

When analyzing 3-phase conditions in transient, time-domain differential equations are used.

5 3-Phase Transformer

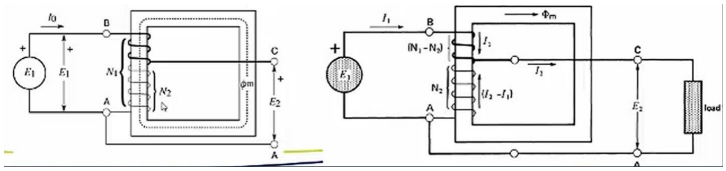
5.1 Dual Voltage Transformer:

These are used to supply power to residental areas. The normal rating of this is 3kVA-500kVA.



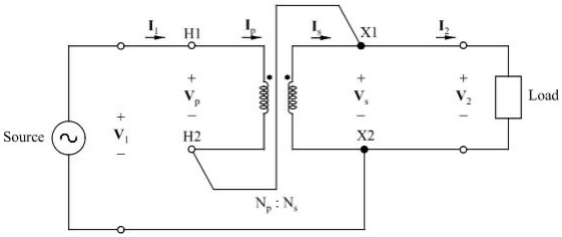
5.2 Auto Transformer:

This has a single winding with taps within the winding to create the turns ratio.



$$I_1(N_1 - N_2) = (I_2 - I_1)N_2$$
$$I_1 N_1 = I_2 N_2$$
$$|S| = E_1 I_1 = E_2 I_2$$

Eliminates a need for a secondary winding. This saves on weight, size and cost. However there is no electrical isolation for the secondary circuit.

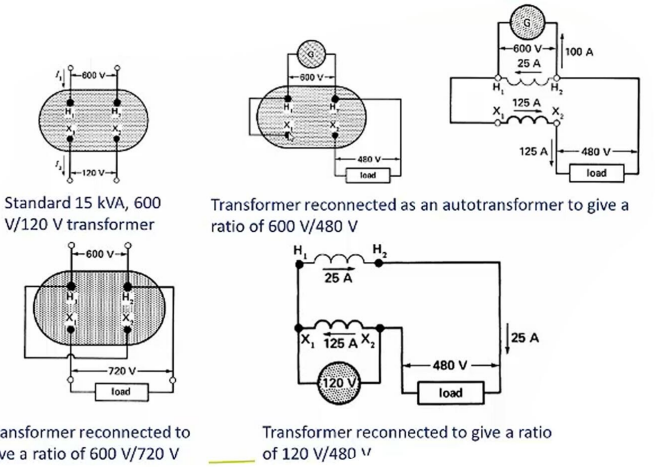


Source Voltage: $V_1 = V_p + V_s$
Source Current: $I_1 = I_p$
Rated Load Voltage: $V_s = V_2$
Rated Load Current: $I_2 = I_p + I_s$
Apparent Load Power: $S_L = V_2 I_2 = V_s(I_p + I_s)$

5.3 Conventional Transformer as an Auto Transformer:

A normal 2 winding transformer can be connected as an auto transformer if:

- Current in any winding does not exceed nominal
- Voltage across windings does not exceed nominal
- Current flow in one winding will automatically flow in another
- Voltage across one winding is automatically across another.

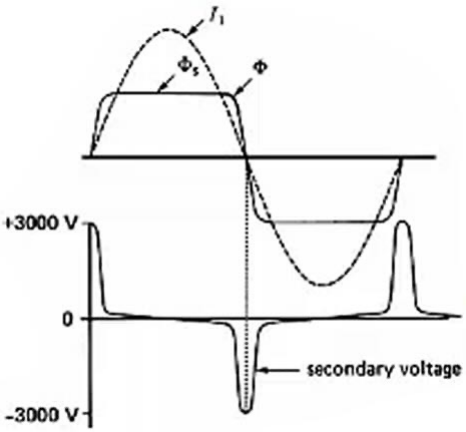


5.4 Voltage Transformer:

- High precision transformer: $P \rightarrow S$ voltage ratio is constant
- Secondary voltage is in phase with the primary
- Usual secondary voltage of 115V
- Usually HV line measure Line-Nuetral

5.5 Current Transformers:

- High precision transformers
- I_p and I_s are functionally in phase
- Usual I_s is SA
- $I_1\ I_2 = N_1\ N_2$
- Do not leave open circuit in current transformers



5.6 Toroidal Current Transformer:

- Used on line carrying more then 100A
- Primary has a single turn
- A laminated ring shaped core cariries the secondary

5.7 Variable Auto-Transformer:

- Used to obtain a variable AC voltage from a fixed AC source
- Has a single layer winding uniformly wound on a toroidal core

5.8 Induction Heating Transformer:

- High power induction furnaces use the transformer principle
- The only difference is that it operates at 500Hz
- Coil is primary whilst molten is secondary
- $|S|$ rating of 15kVA-40MVA

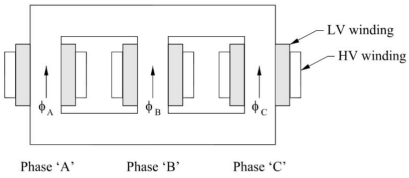
5.9 High Impedance Transformer:

- Some applications need much high leakage reactances (0.9pu) compared to normal (0.03pu)
- Main purpose is to suppress current under short circuit conditions

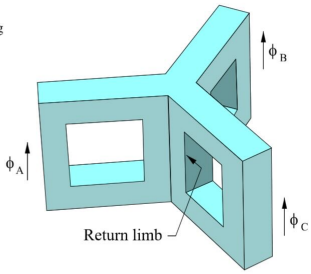
5.10 High Frequency Transformer:

- Electronic PSUs need to isolate output from input
- Operating at frequency of 5kHz-50kHz
- Increase frequency reduces the size of components needed

5.11 3-Phase Transformers Core Config:



Three phase core type
(smaller, lighter and cheaper)

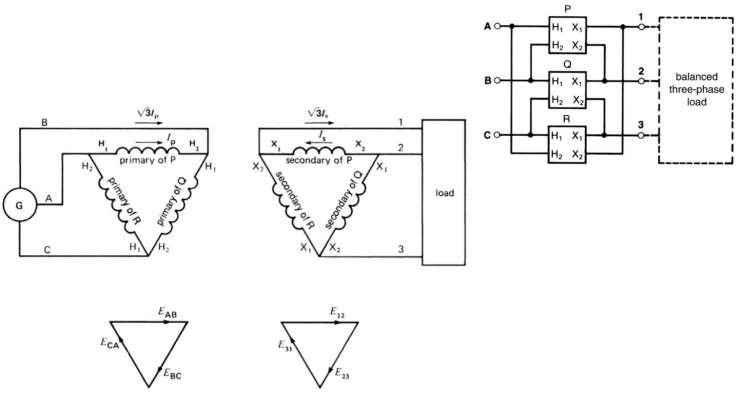


Core with return limb

5.12 3-Phase Connections:

These are number of different connections for 3-phase operation. An important aspect to note is that the connection change between types introduces a phase shift on the signal.

5.12.1 Delta-Delta

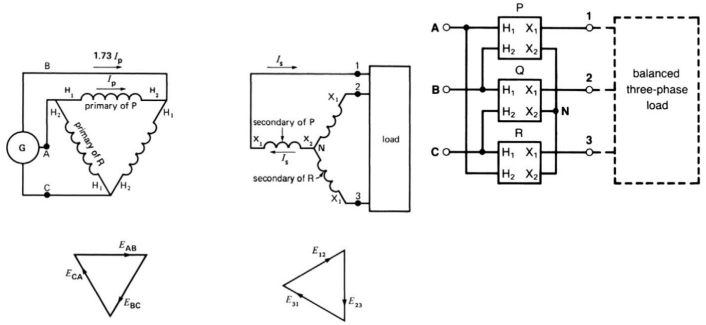


All in phase

$$V_A = V_{AB}$$

$$V_1 = V_{12}$$

5.12.2 Delta-Wye

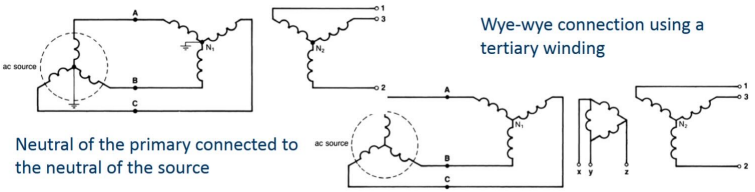


Wye-Delta is functionally the same. Just primary and secondary connections are swapped.

$$V_{AB} = \sqrt{3} V_{AN} \angle 30°$$

$$V_{ab} = \frac{N_2}{\sqrt{3} N_1} V_{AB} \angle -30°$$

5.12.3 Wye-Wye

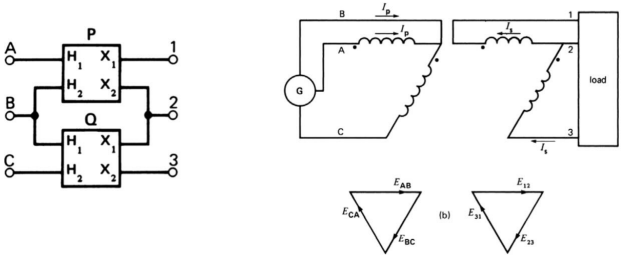


Care must be taken to distant severe distortion of Line-Neutral voltage. Connect nuetral of primary to nuetral of secondary. Provide a tertiary winding, so line voltages are in phase.

$$\frac{V_{an}}{V_{AN}} = \frac{N_2}{N_1}$$

$$V_{AB} = \sqrt{3} V_{AN} \angle 30°$$

5.12.4 Open-Delta Connection



A method to transform 3-phase using 2 transformers. The connection type is identical to Delta-Delta connection. An important notes is the load capacity is 86.6% of the install transformer capacity.

5.13 Vector Group for Transformer Connections:

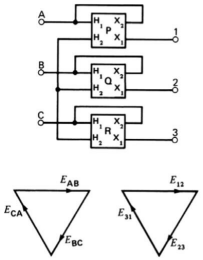
Y - HV Wye Connection
D - HV Delta Connection
y - LV Wye Connection
d - LV Delta Connection
11 - A phase lead of 30° in LV with respect to primary
1 - A phase lag of 30° in LV with respect to primary
0 - No phase difference

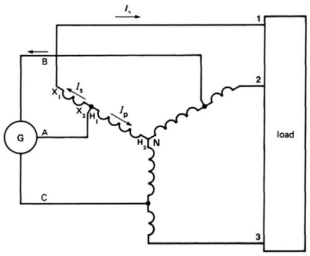
Wye-Wye	Yy0
Delta-Delta	Dd0
Wye-Delta	Yd1
Delta-Wye	Dy11

5.14 ANSI Standard Connection:

When lines are connected in parallel, there can be no phase difference between the voltage lines. In the case of Wye-Delta transformer windings, the connection are always done in a way, so HV quatities lead LV quantities by 30°.

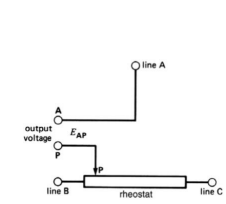
5.15 Special 3-Phase Auto Transformers:

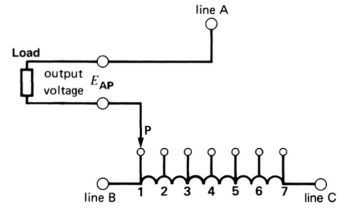




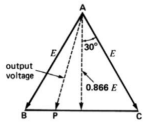
Used when 3-Phase line voltage has to be stepped up or down by a considerable amount. It is cheaper to connect 3-single phase wye connected autotransformers.

5.16 Phase Shift Principle:

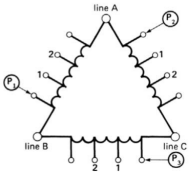




Autotransformer as a phase shifter



Phase shifter using a potentiometer



Three-phase phase shifter

Phase shifting enables creating 2-phase, 6-phase, or even 12-phase systems from an ordinary 3-phase power transfer.

5.17 Analyzing 3-Phase Transformers within Circuits:

- Assume P and S are wye connected (if not derive the equivalent circuit configuration).
- Consider 1-phase of the entire 3-phase system.
- V_p of the transformer must be taken as the L-N of the line.
- V_s is the L-N of the out going line.
- Nominal power rating of the transformer is $\frac{1}{3}$ of the 3-phase transformer bank.
- The load is $\frac{1}{3}$ of the load rating.